The Impact of Dobbs v. Jackson Women's Health Organization on Teen Birth Rates in the United States

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This paper examines whether the overturning of *Roe v. Wade* in Dobbs v. Jackson Women's Health Organization led to an increase in the number of children born to women aged 12–19. The overturning of Roe v. Wade in *Dobbs v. Jackson Women's Health Organization* (2022) represents one of the most significant changes in U.S. reproductive policy in decades. By allowing states to restrict or ban abortion, this policy change may have significantly affected fertility behavior, particularly among teenagers, who face higher risks of unintended pregnancy and greater barriers to accessing reproductive healthcare. Understanding this relationship is important for evaluating the broader economic and social consequences of abortion policy, especially given the long-term impacts of teenage childbearing on education, labor market outcomes, and overall well-being.

Understanding whether abortion restrictions increase births among women aged 12–19 is important for several reasons. First, teenage childbearing is strongly associated with lower educational attainment, reduced labor market participation, and lower lifetime earnings. If abortion restrictions increase teen births, these policies may have long-term economic consequences for affected individuals. Second, this question is relevant for evaluating the broader social and economic impacts of abortion policy. Changes in teen fertility can affect public spending on healthcare, education, and social assistance programs, as well as intergenerational outcomes for children. Finally, this analysis contributes to an ongoing policy debate by providing causal evidence on how legal access to abortion influences real economic and demographic outcomes. Identifying these effects helps policymakers and researchers better understand the trade-offs associated with abortion restrictions.

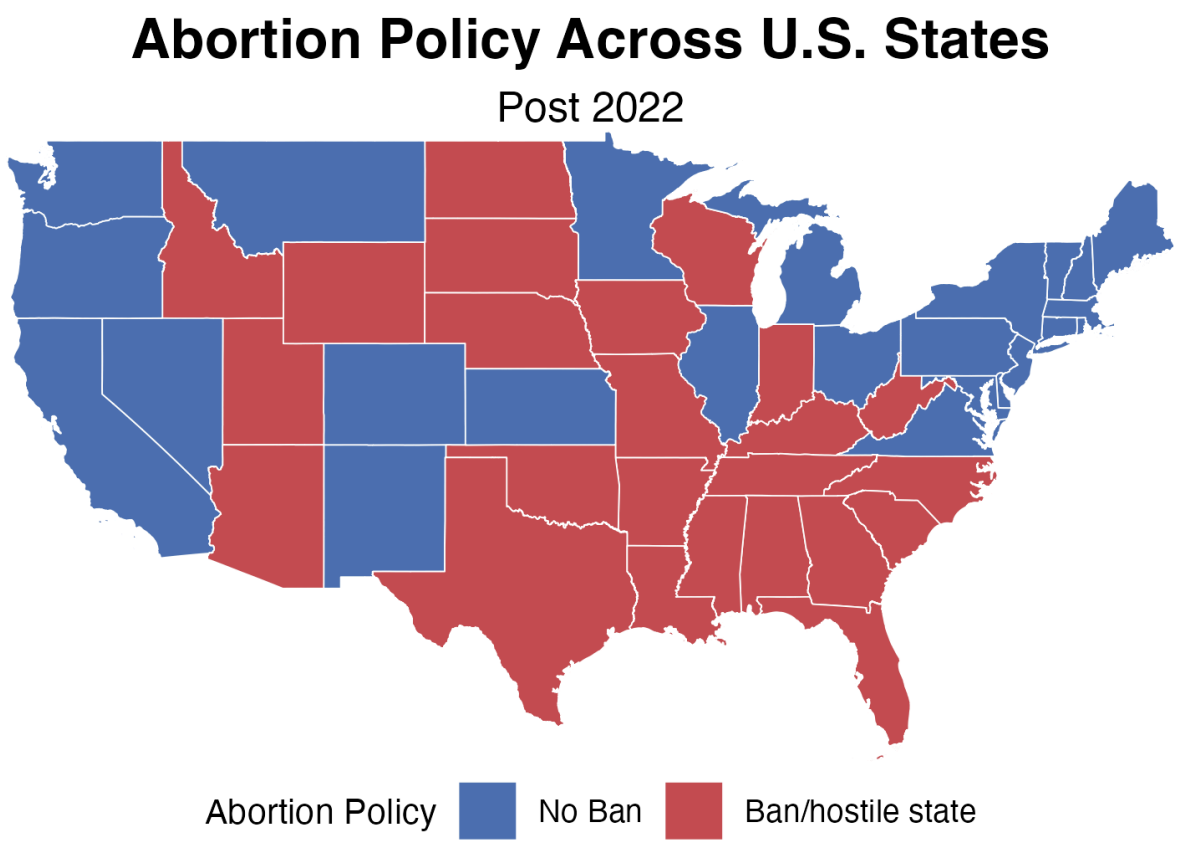
Economic theory predicts that the overturning of Roe v. Wade in Dobbs v. Jackson is expected to increase the number of births to women aged 12–19, in states that implemented strict abortion restrictions. In a standard model of fertility choice, individuals decide whether to have a child by comparing the expected benefits and costs. Access to abortion lowers the cost of avoiding an unintended birth after pregnancy occurs. When abortion becomes restricted, this option is reduced or eliminated, increasing the likelihood that unintended pregnancies result in births. Teenagers are especially sensitive to this change for several reasons. They are more likely to experience unintended pregnancies and typically face higher costs of accessing abortion services, such as financial constraints and limited ability to travel across state lines. As a result, restrictions on abortion access are expected to have a larger effect on fertility among women aged 12–19 compared to older women. From a human capital perspective, early childbearing increases the opportunity cost of education and labor market participation. Without access to abortion, teenagers may be less able to delay childbirth, leading to higher observed birth rates in this group.

Recent economics literature finds that abortion restrictions increase births, with effects depending heavily on access barriers such as distance to providers. In *"The Road Not Taken: How Driving Distance and Appointment Availability Shape the Effects of Abortion Bans,"* Dench, Myers, and Pineda-Torres (2025) use a difference-in-differences design to estimate the

impact of abortion bans on births at the county level. They find that abortion bans lead to an increase in births, but the magnitude depends on how difficult it becomes to access abortion services. Specifically, the authors find that abortion bans increase births even when travel distance to providers remains unchanged, suggesting a baseline "chilling effect" caused by legal uncertainty or logistical barriers. When access becomes more difficult—such as when the nearest provider is much farther away—the effects are substantially larger; for instance, increasing travel distance from 50 miles to 300 miles is associated with roughly a 2–3% rise in births. Limited appointment availability further amplifies these effects. The largest increases in births are observed among disadvantaged groups, including younger, less-educated, and unmarried women, highlighting how restrictions disproportionately affect those with fewer resources to navigate barriers to care.

Related work by the same authors using early post-Dobbs data finds that states implementing abortion bans experienced an average 2.3% increase in births relative to states without restrictions. Overall, the literature suggests that abortion restrictions increase fertility, with stronger effects in areas where access becomes more limited. These findings are consistent with economic theory, which predicts that higher costs of obtaining an abortion lead to more births. This paper builds on this literature by focusing specifically on teenage women (ages 12–19), a group that may be particularly sensitive to changes in access.

I will use a difference-in-differences (DiD) research design that compares changes in teen birth rates across states with different levels of abortion access before and after the policy change. Specifically, I will divide states into two groups: a treatment group consisting of states that implemented strict abortion bans or heavy restrictions after *Dobbs*, and a control group consisting of states that maintained relatively stable abortion access. The key idea is to compare how teen birth rates changed over time in these two groups. If birth rates among teenagers increased more in restrictive states after the policy change (relative to states without restrictions) this difference can be attributed to the effect of abortion restrictions, assuming other factors are similar across groups.



The empirical model would estimate the change in birth rates for women aged 12–19 before and after the *Dobbs* decision. The coefficient of interest is the interaction between a post-*Dobbs* indicator and a treatment-state indicator, which measures the causal effect of abortion restrictions on teen births. The regression equation is as follows:

$$teen_birth_{ist} = \beta_0 + \beta_1\, abortionban_s + \beta_2 post_t + \beta_3(abortionban_s \times post_t) + \epsilon_{ist}$$

The data used in this analysis come from the American Community Survey (ACS) microdata, which provides individual-level information on demographic and socioeconomic characteristics across U.S. states. The unit of observation is a person-year, and the dataset contains over 16 million observations. Key variables include age, sex, state of residence, education, and income, as well as measures related to family structure such as the number of children (nchild) and the number of young children under age five (nchlt5).

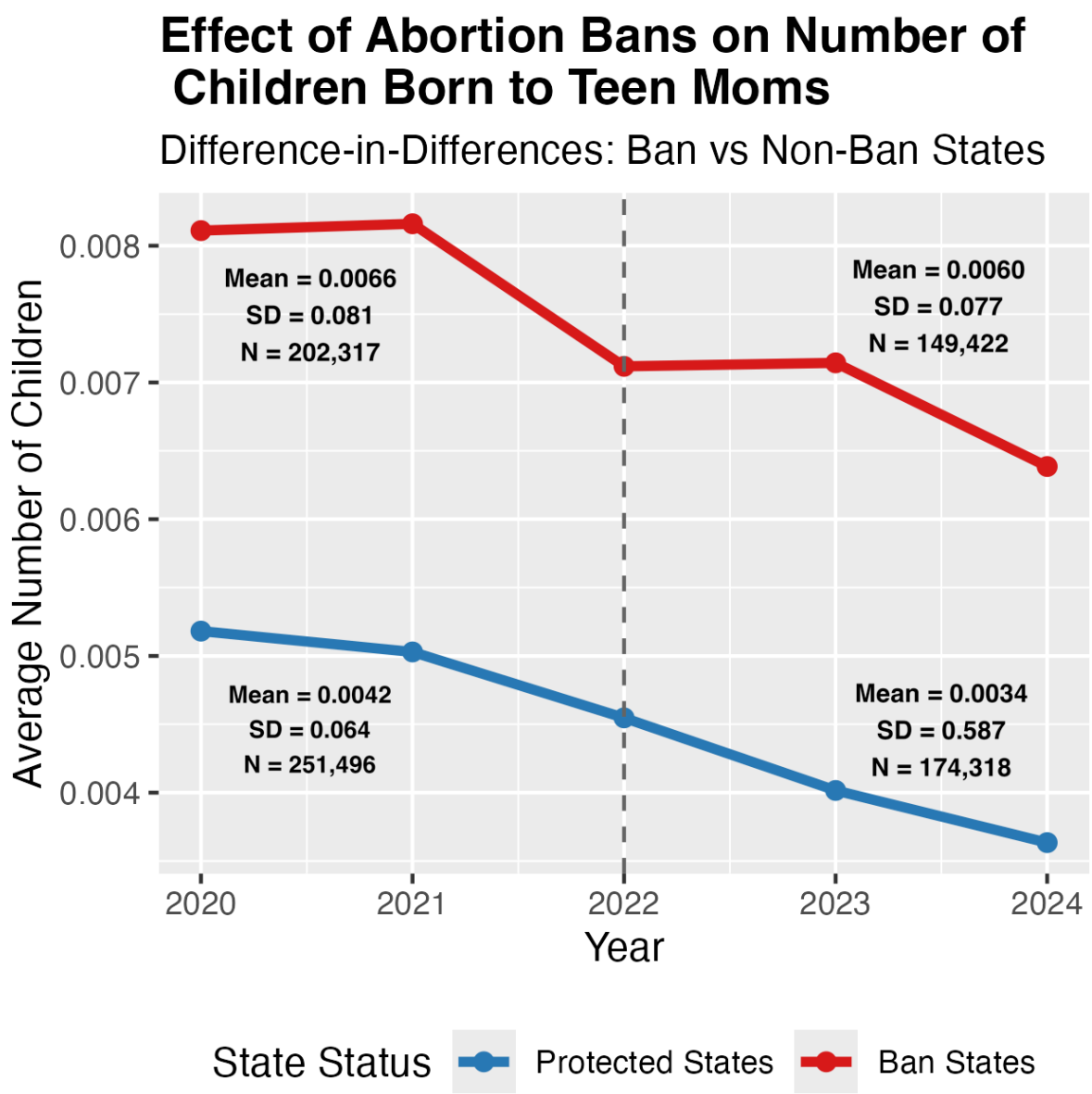
To study teenage fertility, I restrict the sample to females aged 12–19 and construct a proxy for childbirth using indicators for whether a respondent has children. The dataset includes a time variable (year) and state identifiers (statefip), which allow me to exploit variation across states and over time surrounding the Dobbs v. Jackson Women's Health Organization decision. In

addition, the data include socioeconomic controls such as education and income, which help account for factors that may influence fertility decisions.

Table 2: Difference-in-Differences Regression of Teen Births on Abortion Bans

Effect	Estimate	Interpretation
Abortion Ban (β_1)	0.00245***	Higher baseline teen birth rates in ban states
Post-2022 (β_2)	-0.00075***	Teen births declined nationally after 2022
Ban × Post (β_3)	0.00014	Causal effect of abortion bans

*p<0.1; **p<0.05; ***p<0.01



I estimate a difference-in-differences model of teen birth outcomes using an interaction between an abortion ban indicator and a post-Dobbs indicator. The estimated coefficient on the interaction term t is 0.00014. This coefficient is positive, indicating that teen birth rates increased slightly (or declined less) in states that implemented abortion bans after the Dobbs decision, relative to states that did not. The main effects show that states with abortion bans have higher baseline teen birth rates (coefficient = 0.00245), and teen birth rates declined overall after 2022 (coefficient on post = -0.00075) aImportantly, the interaction term isolates the causal effect of abortion restrictions by comparing changes over time across these groups.

The estimated treatment effect of abortion bans on teen birth rates is 0.00014, which corresponds to a 0.014 percentage point increase in the probability that a teenage girl has a child. While this

effect is positive and in the direction predicted by economic theory, it is very small in absolute terms. Given a baseline teen birth rate of approximately 0.4% (0.004), this represents a relative increase of about 3.5%. However, the interaction term is not statistically significant ($p = 0.673$), and the 95% confidence interval includes zero (−0.00051 to 0.00078). This indicates that we cannot reject the null hypothesis that abortion bans had no effect on teen birth rates. Although the point estimate suggests that restrictions may increase teen births, the effect is imprecisely estimated and could be positive, negative, or zero. Overall, teen birth rates declined nationwide after 2022, and while the decline was slightly smaller in states with abortion bans, the difference is minimal and not statistically significant. These findings suggest that the short-run effects of abortion restrictions on teen fertility are limited, and that behavioral adjustments—such as increased contraception use, delayed sexual activity, or travel to access abortion services—may mitigate potential policy impacts.

Several factors could threaten the internal validity of this analysis. First, the difference-in-differences (DiD) method relies on the parallel trends assumption, which requires that, in the absence of the Dobbs decision, teen birth rates in states that implemented abortion bans would have followed the same trajectory as those in states without bans. Based on the ACS data, pre-2022 teen birth rates were slightly higher in treated states (0.0066) than in control states (0.0041), but both groups appear to follow roughly similar trends over time. This suggests that while parallel trends are not grossly violated, there could still be subtle differences in baseline trends that slightly bias the estimates. Second, policy endogeneity may be a concern. States passing abortion bans might systematically differ from other states in ways that also affect teen fertility, such as cultural norms, education, or access to contraception. If these factors change over time in ways that correlate with teen births, the estimated DiD effect may not fully isolate the causal impact of abortion restrictions. Third, the outcome variable itself, teen_birth = (nchild > 0), is an imperfect proxy for recent births. It captures whether a teen ever had a child, not whether she gave birth in the current year, which introduces potential measurement error that can attenuate the estimated effect. Finally, cross-state migration and concurrent policy changes could confound the results. Teens may travel to states without restrictions to obtain abortion services, or other policies affecting fertility (such as changes in contraception access or economic conditions) may influence teen birth rates. If these factors differ across states and are not accounted for, they may bias the DiD estimates.

Even if the estimated effects are internally valid, several factors limit their generalizability. First, the analysis focuses exclusively on women aged 12–19. Older women face different constraints and may respond differently to abortion restrictions, so these results may not generalize to the broader population. Second, the data only capture the immediate post-Dobbs period. Behavioral adjustments, such as increased contraceptive use or delayed sexual activity, may occur over time, meaning the short-run effects observed here might not reflect long-term impacts. Third, geographic variation is an important consideration. The impact of abortion bans likely depends on factors such as distance to the nearest abortion provider and policies in neighboring states.

These effects may vary across regions, limiting the ability to generalize findings nationwide. Finally, the dataset itself has limitations. The ACS is not specifically designed to measure fertility timing, and the proxy for teen births may under- or overstate actual rates. This, combined with small and statistically insignificant estimated effects, means the magnitude of the effect should be interpreted cautiously when applying it to other populations or settings.

References

Dench, D. L., Myers, C. K., & Pineda-Torres, M. (2025). *The road not taken: How driving distance and appointment availability shape the effects of abortion bans* (NBER Working Paper No. 33548). National Bureau of Economic Research. https://www.nber.org/system/files/working_papers/w33548/w33548.pdf

Ruggles, S., Flood, S., Sobek, M., Backman, D., Cooper, G., Rivera Drew, J. A., Richards, S., Rogers, R., Schroeder, J., & Williams, K. C. W. (2025). *IPUMS USA: Version 16.0* [Data set]. IPUMS. https://doi.org/10.18128/D010.V16.0

Posit team (2025). RStudio: Integrated Development Environment for R. Posit Software, PBC, Boston, MA. URL http://www.posit.co/.